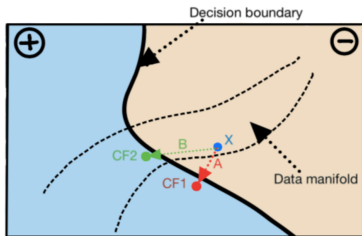
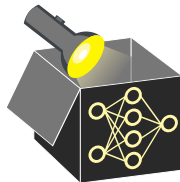


Interpretable Machine Learning

Counterfactual Explanations (CEs)

Motivation



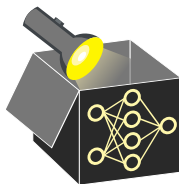
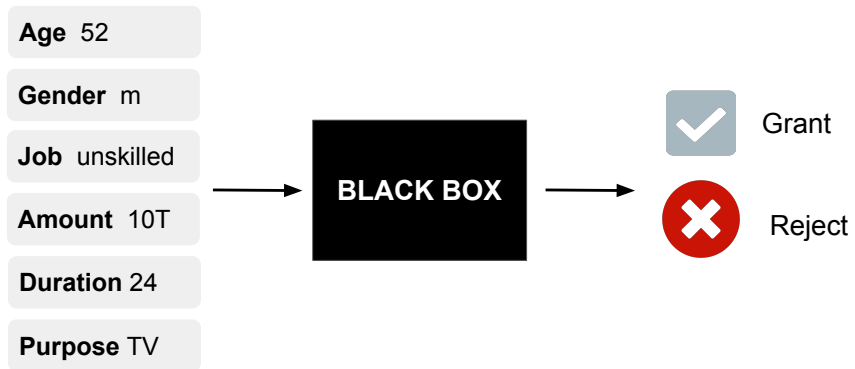
Learning goals

- Understand the motivation behind CEs
- Know why and how CEs are used
- Recognize the philosophical foundations of counterfactual reasoning

MOTIVATING EXAMPLE: CREDIT RISK & CE

x : customer and credit information

y : grant or reject credit

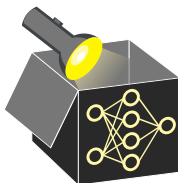
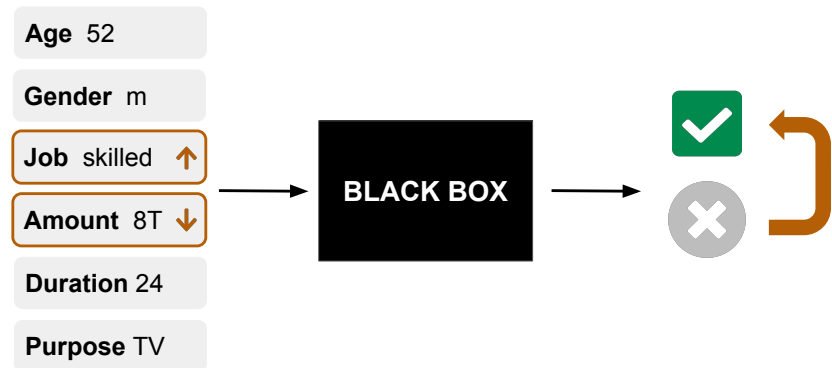


Potential questions:

- Why was the credit rejected?
- Is this decision fair compared with similar applicants?
- **How should x be changed so that the credit is accepted?**

MOTIVATING EXAMPLE: CREDIT RISK & CE

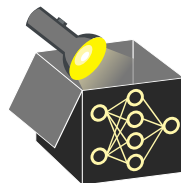
CEs provide answers in the form of “What-If”-scenarios.



“If the applicant had higher skills and the credit amount had been reduced to \$8.000, the loan would have been granted.”

CORE DEFINITION AND PURPOSE OF CE

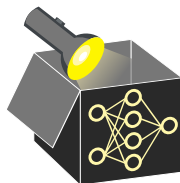
- **Counterfactual explanation (CE):** Hypothetical input $\mathbf{x}'$ close to the data point of interest $\mathbf{x}$ whose prediction equals a user-defined desired outcome y'



CORE DEFINITION AND PURPOSE OF CE

- **Counterfactual explanation (CE):** Hypothetical input $\mathbf{x}'$ close to the data point of interest $\mathbf{x}$ whose prediction equals a user-defined desired outcome y'
- **Proximity constraint:**

Find $\mathbf{x}' \approx \mathbf{x}$ such that $f(\mathbf{x}') = y'$ and distance $d(\mathbf{x}, \mathbf{x}')$ is minimal



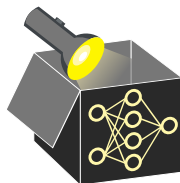
CORE DEFINITION AND PURPOSE OF CE

- **Counterfactual explanation (CE):** Hypothetical input $\mathbf{x}'$ close to the data point of interest $\mathbf{x}$ whose prediction equals a user-defined desired outcome y'

- **Proximity constraint:**

Find $\mathbf{x}' \approx \mathbf{x}$ such that $f(\mathbf{x}') = y'$ and distance $d(\mathbf{x}, \mathbf{x}')$ is minimal

- **Minimal actionable changes:** Difference $\mathbf{x}' - \mathbf{x}$ shows the smallest feature change a user could realize in practice



CORE DEFINITION AND PURPOSE OF CE

- **Counterfactual explanation (CE):** Hypothetical input $\mathbf{x}'$ close to the data point of interest $\mathbf{x}$ whose prediction equals a user-defined desired outcome y'

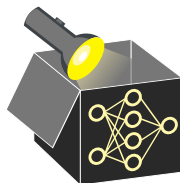
- **Proximity constraint:**

Find $\mathbf{x}' \approx \mathbf{x}$ such that $f(\mathbf{x}') = y'$ and distance $d(\mathbf{x}, \mathbf{x}')$ is minimal

- **Minimal actionable changes:** Difference $\mathbf{x}' - \mathbf{x}$ shows the smallest feature change a user could realize in practice

- **Primary audience:**

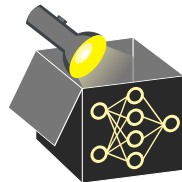
- Individuals aiming to alter model predictions
- ML engineers exploring model behavior under adversarial conditions
 $\rightsquigarrow$ how small text changes in an email flip the prediction from “spam” to “no spam”



INTERPRETIVE AIMS & ROLES

CEs can serve various purposes; the user can decide what to learn from them, e.g.:

“If the person had been **one year older** and the **credit amount had been increased** to \$12.000, the credit would have been granted.”



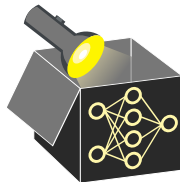
INTERPRETIVE AIMS & ROLES

CEs can serve various purposes; the user can decide what to learn from them, e.g.:

“If the person had been **one year older** and the **credit amount had been increased** to \$12.000, the credit would have been granted.”

- **Guidance for future actions:**

Ok, I will apply again next year for the higher amount.



INTERPRETIVE AIMS & ROLES

CEs can serve various purposes; the user can decide what to learn from them, e.g.:

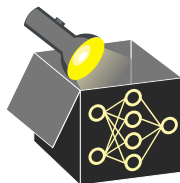
“If the person had been **one year older** and the **credit amount had been increased** to \$12.000, the credit would have been granted.”

- **Guidance for future actions:**

Ok, I will apply again next year for the higher amount.

- **Provide reasons:**

Interesting, I did not know that age plays a role in loan applications.



INTERPRETIVE AIMS & ROLES

CEs can serve various purposes; the user can decide what to learn from them, e.g.:

“If the person had been **one year older** and the **credit amount had been increased** to \$12.000, the credit would have been granted.”

- **Guidance for future actions:**

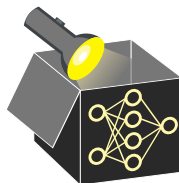
Ok, I will apply again next year for the higher amount.

- **Provide reasons:**

Interesting, I did not know that age plays a role in loan applications.

- **Provide grounds to contest the decision:**

How dare you, I won't accept age-based discrimination in my application.



INTERPRETIVE AIMS & ROLES

CEs can serve various purposes; the user can decide what to learn from them, e.g.:

“If the person had been **one year older** and the **credit amount had been increased** to \$12.000, the credit would have been granted.”

- **Guidance for future actions:**

Ok, I will apply again next year for the higher amount.

- **Provide reasons:**

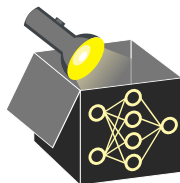
Interesting, I did not know that age plays a role in loan applications.

- **Provide grounds to contest the decision:**

How dare you, I won't accept age-based discrimination in my application.

- **Detect model biases:**

There is a bug, an increase in amount should not increase approval rates.



PHILOSOPHICAL FOUNDATIONS

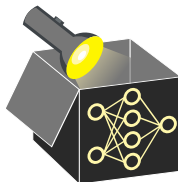
► "Lewis" 1973

Counterfactuals have a long tradition in analytic philosophy

↪ A **counterfactual conditional** takes the form:

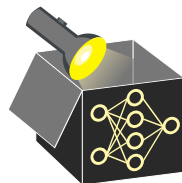
"If S had occurred, Q would have occurred."

- S : past event that never happened
↪ CE run contrary to fact
- Statement is true iff Q holds in all **closest** worlds where S is true
- Closest worlds preserve laws and alter as few facts as possible (related to S)



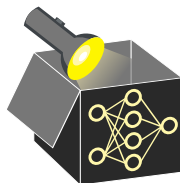
PHILOSOPHICAL FOUNDATIONS

- CEs have largely been studied to explain causal dependence
- **Causal dependence:** Q depends on $S \Leftrightarrow$ without S , no Q
 - ↪ Good CEs reveal critical causal factors that drove algorithmic decision
 - ↪ **CE objective:** find $\mathbf{x}' \approx \mathbf{x}$ with $f(\mathbf{x}') = y'$ to expose causal features



PHILOSOPHICAL FOUNDATIONS

- CEs have largely been studied to explain causal dependence
- **Causal dependence:** Q depends on $S \Leftrightarrow$ without S , no Q
 - ↪ Good CEs reveal critical causal factors that drove algorithmic decision
 - ↪ **CE objective:** find $\mathbf{x}' \approx \mathbf{x}$ with $f(\mathbf{x}') = y'$ to expose causal features
- Relaxing closeness may add causally irrelevant edits to the explanation
 - ↪ e.g., suggest to lower loan *and* increase age (but only loan matters)



PHILOSOPHICAL FOUNDATIONS

- CEs have largely been studied to explain causal dependence
- **Causal dependence:** Q depends on $S \Leftrightarrow$ without S , no Q
 - ↪ Good CEs reveal critical causal factors that drove algorithmic decision
 - ↪ **CE objective:** find $\mathbf{x}' \approx \mathbf{x}$ with $f(\mathbf{x}') = y'$ to expose causal features
- Relaxing closeness may add causally irrelevant edits to the explanation
 - ↪ e.g., suggest to lower loan *and* increase age (but only loan matters)
- CEs are contrastive: Explain a decision by comparing it to a different case
 - ↪ If age were 30 (not 60), loan would have been \$9k (not rejected)
 - ↪ Answers contrastive question:
“Why Q' instead of Q ?” (preferred by humans)

